

Granuloma Inguinale (Donovanosis): Comprehensive Disease Profile

Disease: Granuloma Inguinale (Donovanosis) · **MONDO:** MONDO:0005777 · **Category:** Infectious Disease **Report date:** 2026-09-25 · **Evidence base:** Human clinical studies, case reports/series, phylogenetic and ultrastructural studies, and management guidelines (primarily aggregated disease-level literature, not individual EHR).

Summary (Answer to the Research Question)

Granuloma inguinale (donovanosis) is a **chronic, slowly progressive, mildly contagious sexually transmitted bacterial infection** that produces characteristic **beefy-red, friable, readily bleeding genital/inguinal ulcers**. It is caused by the intracellular Gram-negative bacillus *Klebsiella granulomatis* (formerly *Calymmatobacterium granulomatis*). The organism parasitizes tissue **macrophages/histiocytes**, driving a **granulomatous, plasma-cell-rich inflammatory response** with overlying pseudoepitheliomatous hyperplasia; the intracytoplasmic bacteria are visible as pathognomonic **Donovan bodies**. Diagnosis rests on cytology/histology (Donovan bodies) and, increasingly, PCR, because the organism is difficult to culture. **Azithromycin is first-line therapy**, with cure the norm. It is a **rare, geographically restricted disease** (endemic foci in Papua New Guinea, southern Africa, India, Brazil, and formerly among Aboriginal Australians) that is **declining toward eradication**, and it is an established **cofactor for HIV-1 acquisition/transmission**. As a purely infectious disease it has **no human causal gene, no Mendelian inheritance, and no established animal model**; several template sections (host genetics, model organisms, germline variants) are therefore Not Applicable.

1. Disease Information

- **Overview:** A chronic ulcerative bacterial STI of the genital and inguinal region. Genital papules appear after a ~50-day incubation and evolve into progressively enlarging,

painless, friable ulcers with a beefy-red granulation base that bleed on contact. Course is indolent and progressive without treatment; healing occurs with antibiotics.

• **Key identifiers:**

• **MONDO:** MONDO:0005777

• **ICD-10:** A58 (Granuloma inguinale); **ICD-11:** 1A76 (Granuloma inguinale)

• **MeSH:** D006099 "Granuloma Inguinale"

• **SNOMED CT:** 5006005 (Granuloma inguinale)

• **OMIM / Orphanet:** Not applicable as a Mendelian disorder (infectious disease; no OMIM phenotype entry). Orphanet does not list it as a rare genetic disease.

• **Causative organism NCBI Taxonomy:** *Klebsiella granulomatis* (txid280) / *Calymmatobacterium granulomatis*.

• **Synonyms / alternative names:** Donovanosis; granuloma venereum; granuloma inguinale tropicum; "serpiginous ulcer" (historical); ulcerating granuloma of the pudenda. (Note: distinct from *granuloma inguinale tropicum*/LGV historically confused terminology.)

• **Information source:** Aggregated disease-level literature (guidelines, reviews, clinical case series), not individual-patient EHR.

Evidence:

P 41016613

— "Donovanosis usually causes genital ulcers with a distinct clinical appearance... The causative organism is a gram-negative bacillus, *Calymmatobacterium granulomatis*."

P 42486240

(historical review).

2. Etiology

- **Causal factor — infectious:** *Klebsiella granulomatis* (formerly *Calymmatobacterium granulomatis*), an encapsulated, intracellular **Gram-negative bacillus** of family Enterobacteriaceae. Transmission is predominantly **sexual** (genital, and via ano-receptive/ oro-genital contact); low infectivity, with autoinoculation contributing to spread. There is **no genetic/heritable cause**.

- **Risk factors (environmental/behavioral):** residence in or travel to endemic areas; multiple sexual partners; low socioeconomic status/poor access to care; male sex and young adult age (peak in sexually active adults, ~15–40 y); poor genital hygiene; possibly fecal contamination/anal exposure. **No genetic susceptibility loci** have been identified.
- **Protective factors:** condom use, partner notification/treatment, sexual abstinence during treatment, early antibiotic therapy, and general STI-control/health-education programs. No genetic protective variants (Not Applicable).
- **Gene–environment interactions:** Not applicable to host genetics. The clinically relevant interaction is **comorbidity with HIV**, where immunodeficiency prolongs/worsens disease and requires extended therapy.

Evidence:

P 10555350

P 10482295

P 22239475

— reclassification and phylogeny;

P 41016613

— Gram-negative bacillus, nomenclature debate.

3. Phenotypes

Primary phenotype — chronic genital ulcer (HP:0000163 region; HP:0200035 "Genital ulcers"): - *Type:* clinical sign / physical manifestation (mucocutaneous ulceration). - *Characteristics:* adult-onset; typically **painless or minimally painful; beefy-red, granulomatous, friable base that bleeds readily on contact;** slowly **progressive/enlarging;** chronic (weeks–months). In a Durban series, ulcers persisted >28 days in ~55% of men and ~46% of women. - *Four recognized morphologic subtypes:* **ulcerogranulomatous** (most common), **hypertrophic/verrucous, necrotic** (deep, foul-smelling, destructive), and **sclerotic/cicatricial** (fibrotic).

Associated signs/complications: - **Pseudobuboes** — subcutaneous inguinal granulomas mimicking lymphadenopathy (HP:0002716 lymphadenopathy — note: true lymphadenitis is usually *absent*). - **Genital/vulval swelling** → **elephantiasis** (HP:0100820) from lymphatic obstruction (chronic cases). - **Tissue destruction/mutilation, scarring, stricture**; genital lymphedema. - **Secondary squamous cell carcinoma** (HP:0002860) in long-standing lesions (rare).

Frequency: Ulcerogranulomatous form predominates (>90% of lesions in series). **Quality-of-life impact:** substantial physical disability, disfigurement, sexual/urinary dysfunction, and psychological distress/stigma, particularly with elephantiasis or mutilating disease; formal EQ-5D/SF-36 data are not available for this rare disease. Stigma is amplified by sensationalist media: reports of a "flesh-eating infection donovanosis" are false and "only leading to hyperbole and increased stigma among those infected" (

P 41016613

Pregnancy & special populations: In a Durban series of 123 women, 42% were pregnant; in ~85% donovanosis had no effect on pregnancy outcome and there was **no evidence of congenital disease in neonates** (

P 8735293

though extensive vulval lesions can complicate delivery. **Erythromycin** is the preferred agent in pregnancy. Disease behaves similarly in HIV-positive and HIV-negative women, though healing may be slower with advanced immunodeficiency.

Evidence:

P 26882914

21097731 — incubation and lesion types;

P 8509089

— lesion-type frequencies and chronicity;

P 16510000

— elephantiasis.

4. Genetic / Molecular Information

Not applicable (host). Donovanosis is an infectious disease with **no causal human gene, no pathogenic germline/somatic variant, no modifier genes, no disease-defining epigenetic changes, and no chromosomal abnormalities.** There is no inheritance, penetrance, expressivity, or carrier frequency.

Pathogen molecular biology (limited): *K. granulomatis* shows ~95% 16S rRNA identity to *Klebsiella* and ~94% to *Enterobacter*; sequencing of 16S rRNA + *phoE* (2089 bp) supported reclassification as *Klebsiella granulomatis* comb. nov. Because the organism is **fastidious and hard to culture**, its genome and virulence factors remain **poorly characterized**; a defined virulence repertoire is not established.

Evidence:

P 10555350

P 10482295

P 22239475

— "Because of the difficulty in growing this bacterium... its characteristics have not been sufficiently defined."

5. Environmental Information

- **Environmental/occupational toxins:** none implicated.
- **Lifestyle factors:** high-risk sexual behavior; multiple partners; migration between urban/rural endemic areas (identified as a driver of transmission and HIV co-risk in South African studies). Minimal condom use in affected populations.
- **Infectious agent:** *Klebsiella (Calymmatobacterium) granulomatis* — Gram-negative bacillus; NCBI Taxonomy *Klebsiella granulomatis*; gamma-Proteobacteria; Enterobacteriaceae. Humans are the only known host/reservoir; no environmental or animal reservoir is established.

Evidence:

P 1398660

— sexual-behavior/migration patterns;

P 10482295

— taxonomy.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical manifestation)

1. **Sexual/auto-inoculation exposure** deposits *K. granulomatis* onto genital/inguinal epithelium (often at microabrasions) → **leads to** local colonization. (*demonstrated: sexual transmission; exact portal inferred*)
2. Bacteria are **phagocytosed by tissue macrophages/histiocytes** and **survive/replicate within cytoplasmic vacuoles (phagosomes)** → **results in** an intracellular reservoir (Donovan bodies). (*demonstrated by ultrastructure*)
3. Persisting intracellular bacteria **drive chronic granulomatous inflammation** — recruitment of **plasma cells, neutrophils, and activated macrophages** with few lymphocytes → **leads to** granulation-tissue formation. (*demonstrated histologically*)
4. Ongoing inflammation and proteolysis **erode the epithelium** and induce **pseudoepitheliomatous hyperplasia** at ulcer margins → **results in** the characteristic beefy-red, friable, bleeding ulcer that enlarges peripherally. (*demonstrated*)
5. **Branch A (chronic fibrosis):** persistent inflammation **traps and constricts lymphatics** → **leads to** lymphatic obstruction → **genital elephantiasis / lymphedema** and cicatricial/sclerotic scarring, strictures. (*demonstrated mechanism,*

P 16510000

6. **Branch B (neoplasia):** decades-long chronic ulceration and epithelial hyperplasia **predispose to malignant transformation** → **squamous cell carcinoma**. (*inferred from case reports*)

7. **Branch C (HIV facilitation):** the friable, readily bleeding ulcer **disrupts the mucosal barrier** → **facilitates HIV-1 entry/transmission** (portal of entry + blood contact). *(epidemiologically demonstrated)*
8. **Branch D (spread):** direct extension/autoinoculation and (rarely) hematogenous/lymphatic dissemination → **extragenital, pelvic, or disseminated lesions**. *(demonstrated in case reports)*

Detail by category

- **Cellular processes:** chronic inflammation (GO:0006954), phagocytosis (GO:0006909), granuloma formation; macrophage activation (increased lysosomes, rough ER, filopodia on ultrastructure).
- **Immune involvement:** predominantly **innate/macrophage-centered** with prominent **plasma-cell (humoral) infiltrate**; sparse lymphocytes; the organism resists intracellular killing, enabling persistence. Immunodeficiency (HIV) worsens/prolongs disease.
- **Tissue damage mechanisms:** inflammatory tissue destruction, ulceration, fibrosis/sclerosis, lymphatic obstruction; **not** toxin- or ischemia-driven.
- **Molecular pathways / metabolic / epigenetic / omics:** **Not characterized** — no transcriptomic, proteomic, metabolomic, or defined signaling-pathway data exist for donovanosis (knowledge gap due to culture difficulty).
- **Cell types (CL):** macrophage (CL:0000235), plasma cell (CL:0000786), neutrophil (CL:0000775). **GO cellular component:** phagocytic vesicle/phagosome (GO:0045335).

Evidence:

P 9856642

(intramacrophage localization);

P 2777345

(ultrastructure of inflammatory response);

P 18317211

(plasma-cell/histiocyte histology, Donovan bodies);

P 16510000

(lymphatic constriction → elephantiasis);

P 2063236

(HIV facilitation).

7. Anatomical Structures Affected

- **Organ/system level: reproductive/genitourinary system and integumentary system (skin)** are primary. Secondary: inguinal soft tissue, lymphatics (obstruction), and rarely pelvis/bone (dissemination).
- **Primary sites (~90%):** external genitalia and inguinal/perineal region — penis, prepuce, coronal sulcus, glans; vulva, labia, fourchette; also **uterine cervix** (UBERON:0000002) and **vagina** (UBERON:0000996). **Perianal/anal** lesions with ano-receptive intercourse.
- **Extragenital (rare):** lip/oral cavity, and skin sites such as the foot; disseminated pelvic/visceral disease historically reported.
- **Tissue/cell level:** epithelial tissue (pseudoepitheliomatous hyperplasia) and dermal connective/granulation tissue; key cells = macrophages/histiocytes (CL:0000235), plasma cells (CL:0000786), neutrophils (CL:0000775).
- **Subcellular:** bacteria within macrophage **cytoplasmic vacuoles/phagosomes** (GO:0045335).
- **Localization/lateralization:** lesions localize to sites of sexual contact; inguinal involvement may be **unilateral or bilateral**; "kissing lesions" from autoinoculation on apposed skin.
- **UBERON suggestions:** external genitalia; UBERON:0000996 (vagina); UBERON:0000002 (cervix); UBERON:0002097 (skin of body); inguinal region.

Evidence:

P 8735293

(rectal/pelvic	+	genital);
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P 9924476

(extragenital
foot);

P 18317211

(cervix).

8. Temporal Development

- **Onset: adult-onset**, in sexually active individuals; **insidious/chronic** onset after an **incubation period of ~50 days** (reported range ~1–12 weeks, up to months). Not congenital; neonatal/perinatal transmission is rare.
- **Progression:** untreated disease is **slowly progressive** with peripheral enlargement and local tissue destruction over months–years; stages run from **papule → early ulcer → extensive ulcerogranulomatous/hypertrophic disease → sclerotic/cicatricial end-stage** with scarring, lymphedema/elephantiasis, and rare malignant transformation.
- **Course pattern:** chronic progressive; **not** self-limiting. **Relapse** can occur 6–18 months after apparently adequate therapy, warranting follow-up.
- **Remission: treatment-induced** (antibiotics) — healing from ulcer margins; spontaneous remission is uncommon.
- **Critical intervention window:** early antibiotic treatment prevents tissue destruction, scarring, elephantiasis, and reduces HIV-cofactor risk.

Evidence:

P 26882914

(incubation ~50 days; treat until healed);

P 8509089

(chronicity).

9. Inheritance and Population

- **Epidemiology: rare and geographically restricted**; endemic "hot spots" — **Papua New Guinea, South Africa (KwaZulu-Natal/Durban epidemic 1988–97), India, Brazil, and formerly Aboriginal communities in Australia**. Marked **global decline**; described in 2026 as "well on the way to being eradicated." Precise global incidence/prevalence figures are not maintained in standard registries (SEER/GBD do not track it separately); it is now seen mostly as sporadic imported cases in non-endemic countries.

- **Infectivity: low/mildly infectious** — only 1/21 regular partners were infected in one Durban series, supporting feasibility of elimination programs.
- **Inheritance/penetrance/carrier frequency/founder effects/consanguinity: Not Applicable** (infectious, non-genetic).
- **Demographics:** affects sexually active adults, roughly **15–40 years** (series mean age ~22 in women). **Sex ratio** varies by setting; classic literature reports a **male predominance** (e.g., 130 men vs 41 women in one Durban series), though other series (antenatal/gynecology clinics) are female-predominant, reflecting ascertainment. Higher burden in **lower-income, rural, and marginalized populations** in endemic regions.
- **Geographic variant distribution:** Not characterized (limited pathogen genomics).

Evidence:

P 41016613

("significant global decline... well on the way to being eradicated");

P 26882914

(endemic countries);

P 7750949

(unique geography);

P 8509089

(sex distribution, low partner infectivity).

10. Diagnostics

- **Cornerstone — cytology/histopathology:** microscopic identification of **Donovan bodies** (intracytoplasmic, bipolar "safety-pin" bacteria within macrophages/histiocytes) on **crush/tissue smears** stained with **Giemsa/Wright/RapiDiff** or on biopsy. Histology: granulation tissue with dense plasma-cell infiltrate, neutrophil microabscesses, enlarged vacuolated histiocytes containing Donovan bodies, and pseudoepitheliomatous hyperplasia.
- **Molecular: PCR** (including colorimetric-detection assays) developed for genital-ulcer specimens; increases sensitivity/specificity where available. **LOINC/SNOMED** codes exist for genital-ulcer pathogen testing.

- **Culture:** difficult/not routine (fastidious; grown historically in human peripheral-blood monocyte co-culture and Hep-2 cells).
- **Biomarkers/imaging/electrophysiology/omics:** no specific serum biomarker; imaging only for complications (deep/pelvic/bone extension); no omics-based diagnostics.
- **Genetic testing: Not Applicable** (no host genetic component; WGS/WES/panels/karyotype/CMA/FISH not indicated).
- **Clinical criteria & differential diagnosis:** diagnosis is clinical + cytologic/PCR confirmation. **Differential:** primary syphilis (chancre), chancroid (*H. ducreyi*), lymphogranuloma venereum (*C. trachomatis* L1–L3), genital herpes, secondary bacterial/fungal infection; and non-infectious mimics — **squamous cell carcinoma**, Behçet disease, aphthosis, psoriasis, fixed drug eruption, sexual trauma. Distinguishing features of donovanosis: chronic, painless, beefy-red, friable, bleeding ulcer usually **without true regional lymphadenitis**. *No pathogen is identified in up to 25% of genital-ulcer patients.*
- **Screening:** no asymptomatic/newborn/carrier screening; case-finding via STI services and partner notification.

Evidence:

P 26882914

(Donovan bodies + PCR);

P 22335265

(genital-ulcer differential; up to 25% undiagnosed);

P 39566736

(non-infectious mimics);

P 18317211

(histology).

11. Outcome / Prognosis

- **Prognosis with treatment: excellent** — highly curable with antibiotics; ulcers re-epithelialize over weeks. Early treatment prevents disfigurement and complications.

- **Mortality: very low**; donovanosis is rarely directly fatal. No standardized survival statistics (non-lethal, rare disease). Disease-specific mortality is negligible except via complications (secondary infection, or SCC).
- **Morbidity/disability**: without treatment — extensive genital ulceration, tissue destruction/mutilation, scarring, urethral/vaginal/anal strictures, **genital elephantiasis/lymphedema**, and psychosexual morbidity. Long-standing lesions carry a small risk of **squamous cell carcinoma**.
- **Recovery**: high recovery potential with therapy; established fibrosis/elephantiasis and tissue loss may be **irreversible** and require surgery.
- **Prognostic factors**: duration/extent of lesions at presentation, **HIV coinfection** (prolonged healing, treatment failure risk), and treatment adherence. No molecular prognostic biomarkers.

Evidence:

P 26882914				
(treat		until		healed);
P 7750949				
(treatment	failure	in	advanced	HIV);
P 19061590				
24554002	/	26396449	(SCC	complication);
P 16510000				
(elephantiasis).				

12. Treatment

- **First-line pharmacotherapy: Azithromycin** (macrolide; inhibits 50S ribosome) **1 g weekly** (or 500 mg daily) **continued until all lesions have fully healed** (minimum ~3 weeks). NCIT:C734; CHEBI:2955; ATC J01FA10.
- **Alternative regimens (≥ 3 weeks / until healed):**
- **Doxycycline** 100 mg twice daily (tetracycline) — genital-ulcer guidance specifies **21 days**. NCIT:C744; CHEBI:50845.

- **Trimethoprim–sulfamethoxazole** (co-trimoxazole) 160/800 mg twice daily. NCIT:C265.
- **Erythromycin** 500 mg four times daily (preferred in pregnancy). NCIT:C61780.
- **Ceftriaxone** and **ciprofloxacin** reported as further alternatives.
- **Adjuncts:** add a parenteral aminoglycoside (e.g., **gentamicin**) if lesions do not respond, especially in HIV-positive patients; **prolonged therapy** in advanced HIV.
- **Surgery/interventional:** reserved for complications — excision of fibrotic/mutilating tissue, correction of strictures/elephantiasis, or resection of secondary SCC.
- **Supportive care:** wound care, analgesia, treatment of secondary infection; partner evaluation/treatment.
- **Advanced/experimental therapeutics (gene/cell/RNA/targeted/immuno-therapy) and pharmacogenomics: Not Applicable** — no gene-guided therapy; standard antibiotics suffice. No active registered clinical trials specific to donovanosis.
- **Treatment outcome:** high response/cure rates; monitor for **relapse at 6–18 months**. Adverse events are drug-class-typical (GI upset with macrolides/tetracyclines; photosensitivity with doxycycline; sulfa hypersensitivity).

Evidence:

P 26882914		
21097731	(azithromycin	first-line);
P 22335265		
(doxycycline 21 days).		

13. Prevention

- **Primary prevention:** safer-sex practices (**condoms**), reduction in number of partners, health education, and **prompt treatment of index cases and sexual partners** to interrupt transmission. **No vaccine exists** (immunization Not Applicable).
- **Secondary prevention:** early detection/treatment in STI clinics; **syndromic management** of genital ulcer disease in endemic settings (adapted locally); **partner notification/treatment**; abstinence until lesions heal.

- **Tertiary prevention:** complete antibiotic courses to prevent tissue destruction, strictures, elephantiasis, and malignant transformation; surveillance for relapse and SCC; management of HIV coinfection.
- **Public-health interventions: community-based elimination programs** are feasible given low infectivity and restricted geography — the **Australian Aboriginal donovanosis elimination program** is a documented success and proposed model; donovanosis eradication is framed as an opportunity to also curb HIV-1 spread.
- **Genetic counseling/carrier screening: Not Applicable.**

Evidence:

P 12473810

(successful Australian elimination program; syndromic management);

P 7750949

(global eradication opportunity linked to HIV control).

14. Other Species / Natural Disease

- **Host range: Humans are the only known natural host;** no established zoonotic reservoir or naturally occurring animal disease is documented for *K. granulomatis*.
- **Taxonomy of pathogen:** *Klebsiella granulomatis* (NCBI Taxonomy). Closely related human pathogens in the genus: *K. pneumoniae*, *K. rhinoscleromatis* (causes rhinoscleroma), *K. oxytoca*.
- **Comparative biology:** mechanistic parallels with **rhinoscleroma** (*K. rhinoscleromatis*) — another chronic granulomatous *Klebsiella* infection with intramacrophage bacteria (Mikulicz cells analogous to Donovan-body-laden histiocytes).
- **Breeds/orthologous genes/zoonosis/cross-species susceptibility: Not Applicable / Not documented.**

Evidence:

P 10555350

(relatedness to *K. pneumoniae*, *K. rhinoscleromatis*).

15. Model Organisms

- **No established animal or genetic disease model exists.** The organism's **fastidious growth requirements** (only re-cultured after >30 years in **human peripheral-blood monocyte co-culture and Hep-2 cell** systems) have precluded standard mouse/rat/zebrafish/invertebrate models, and there are no knockout/transgenic/humanized models (host genetics is irrelevant).
- **In vitro systems used:** human monocyte co-cultures and Hep-2 cells for isolation and ultrastructural study.
- **Limitations / research gap:** absence of tractable models and a complete genome is the principal barrier to studying virulence factors, immune evasion, and pathway-level mechanisms.

Evidence:

P 9856642

(monocyte co-culture / tissue ultrastructure);

P 22239475

(culture difficulty; "more studies needed to understand bacterial genetics").

Supported vs Refuted Hypotheses

Supported (evidence-backed): 1. Donovanosis is caused by intracellular Gram-negative *Klebsiella* (*Calymmatobacterium*) *granulomatis* (16S/phoE phylogeny). 2. Pathology is macrophage-based granulomatous inflammation with Donovan bodies (ultrastructure/histology). 3. Azithromycin is first-line, curative therapy (guidelines). 4. Disease is rare, geographically restricted, and declining toward eradication (reviews). 5. Donovanosis is a cofactor for HIV-1 acquisition/transmission (P=0.02 in men; risk rises with lesion duration). 6. Chronic disease causes elephantiasis (lymphatic obstruction) and rarely SCC (mechanism + case reports).

Refuted / Not Applicable: - No host causal gene, inheritance, penetrance, or carrier frequency (infectious disease). - No established animal model or knockout/transgenic system. - No omics/pathway-level molecular profiling; no vaccine; no gene/cell/RNA therapy.

Mechanistic Model (Synthesis)

Sexual / auto-inoculation of *K. granulomatis* (~50-day incubation)



Phagocytosis by dermal MACROPHAGES (CL:0000235)



Intracellular survival in cytoplasmic vacuoles (G0:0045335)
→ DONOVAN BODIES (pathognomonic)



Chronic GRANULOMATOUS inflammation
(plasma cells CL:0000786 + neutrophils CL:0000775)
+ pseudoepitheliomatous hyperplasia



Beefy-red, FRIABLE, bleeding ULCER (HP:0200035)



Lymphatic
constriction
→ ELEPHANTIASIS



Long-standing
lesion →
SCC (rare)



Bleeding portal
→ HIV-1 acquisition
& transmission (cofactor)

([https://pubmed.ncbi.nlm.nih.gov/16510000/](https://pubmed.ncbi.nlm.nih.gov/16510000/ "Visit PubMed"))

The unifying theme is that donovanosis is a **macrophage-parasitizing intracellular infection** whose clinical severity flows from the *chronicity* of the granulomatous response rather than acute toxicity. This explains the slow tempo, the friable vascular ulcers, the late fibrotic/lymphatic and neoplastic complications, the HIV-cofactor role, and why a single mechanistic intervention — sustained intracellular-active antibiotic therapy (azithromycin) — is curative and why elimination programs succeed (breaking the sole human transmission chain).

Evidence Base — Key Papers and How They Support the Findings

PMID	Paper (abbrev.)	Type	Supports
10555350	Reclassification as <i>K. granulomatis</i> comb. nov.	Phylogenetic	Etiology/taxonomy; relatedness to <i>K. pneumoniae</i> / <i>K. rhinoscleromatis</i>
10482295	16S rRNA phylogeny of <i>C. granulomatis</i>	Phylogenetic	~95% <i>Klebsiella</i> , ~94% <i>Enterobacter</i> ; distinct gamma-proteobacterium
22239475	Evolution of STI bacteria	Computational/ review	Culture difficulty; sparse pathogen genetics (knowledge gap)
41016613	Donovanosis review (2026)	Review	Gram-negative bacillus; near-eradication; stigma/misinformation
26882914	2016 European guideline	Guideline	~50-day incubation; 4 lesion types; Donovan bodies/PCR; azithromycin
21097731	2010 European guideline	Guideline	Diagnosis and first-line azithromycin
8509089	Durban clinico-epidemiological study	Clinical series	Lesion-type frequencies; chronicity; sex distribution
2063236	HIV-1 in Durban STD clinic	Cross-sectional (human)	Donovanosis–HIV-1 association (P=0.02, men); risk ↑ with lesion duration
7750949	Global eradication of donovanosis	Review	Unique geography; HIV risk factor; eradication opportunity
9856642	Ultrastructure: culture vs biopsy	In vitro/ ultrastructural	Intramacrophage localization; absent surface structures
2777345	Ultrastructural study of donovanosis	Ultrastructural	Macrophage activation; inflammatory cell repertoire
18317211	Malacoplakia & GI of cervix in AIDS	Case report	Histology (Donovan bodies); cervical involvement
16510000	Genital elephantiasis & STIs	Review	Lymphatic constriction → elephantiasis mechanism
19061590	Malignant transformation (HIV+)	Case report	SCC complication of chronic donovanosis

PMID	Paper (abbrev.)	Type	Supports
8735293	GI in pregnancy & HIV	Clinical series	Benign pregnancy outcome; no congenital transmission
9924476	Extragenital donovanosis of foot	Case report	Rare extragenital localization
22335265	Diagnosis/management of genital ulcers	Review	Doxycycline 21 d; differential diagnosis; ≤25% undiagnosed
39566736	AEDV ulcerative-STI management	Guideline	Non-infectious mimics in differential
12473810	Donovanosis review	Review	Australian elimination program; declining incidence
11394976	Donovanosis: an update	Review	Re-culture after >30 y; prolonged therapy in HIV; cost of azithromycin

Consistency of evidence: Findings are internally consistent across independent guidelines (European 2010/2016), large clinical series (Durban), and mechanistic ultrastructural studies. Taxonomic reclassification is supported by two independent sequencing studies; the HIV-cofactor role by a large cross-sectional study with statistically significant associations. The main evidentiary weakness is reliance on case reports/series for complication rates (SCC, elephantiasis) and the near-total absence of pathogen omics data.

Limitations & Future Directions

- The pathogen's culture difficulty leaves its **genome, virulence factors, and molecular immunology poorly defined**; whole-genome sequencing and in vitro/organoid infection models are key future needs.
- **Epidemiologic surveillance is sparse** (no dedicated global registry); true incidence/prevalence and sex/age distributions are uncertain and setting-dependent.
- Modern PCR-based diagnostics and controlled treatment trials are limited by the disease's rarity.

- Evidence is dominated by case reports/series and expert-guideline consensus rather than RCTs.
- Several template sections (host genetics, model organisms, omics, pharmacogenomics) are **inherently Not Applicable** to a non-genetic infectious disease.

Proposed Follow-up Actions

1. **Whole-genome sequencing of *K. granulomatis*** from clinical isolates (via monocyte/Hep-2 co-culture) to define virulence factors, capsule/LPS loci, and confirm genome-scale *Klebsiella* placement.
 2. **Develop a tractable infection model** (humanized macrophage system, skin organoid, or small-animal model) to study intracellular persistence, Donovan-body formation, and antibiotic penetration.
 3. **Standardized PCR surveillance** in remaining endemic foci (PNG, India, Brazil) to quantify true incidence and monitor progress toward eradication.
 4. **Prospective cohort follow-up** to estimate the risk and triggers of squamous-cell-carcinoma transformation, currently known only anecdotally.
 5. **Integrate donovanosis elimination with HIV-prevention programs**, leveraging the demonstrated HIV-cofactor relationship
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2063236

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 6. **Cost-effectiveness analysis of azithromycin** to support universal first-line adoption (historical cost barrier;

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11394976

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 7. **Public-health communication** to counter "flesh-eating infection" misinformation and reduce stigma
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41016613

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Key Citations (PMID)

41016613 · 42486240 · 10555350 · 10482295 · 22239475 · 12635932 · 26882914 · 21097731 ·
12473810 · 11394976 · 8509089 · 2063236 · 7750949 · 1398660 · 8735293 · 16510000 · 18317211 ·
9856642 · 2777345 · 11100808 · 19061590 · 24554002 · 26396449 · 9924476 · 22335265 · 39566736

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